Data Sheet

EXPERIMENT NO. 1(b)

Title: - Study of Ripple Factor and Voltage Regulation Characteristics of Full-Wave Rectifier With Capacitor Filter (using Tinkercad software circuit).

Apparatus Required: -

Sl. No.	Apparatus / Component	Specification	Quantity

Important Formulae: -

Ripple factor (Theoretical) $\gamma = \frac{1}{4\sqrt{3}fCR}$

$$V_{ac} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

Ripple Factor $\gamma = \frac{V_{ac}}{V_{dc}}$

Observation Table: -

Without Capacitor Filter: -

C(μF)	R_L(KΩ)	V_{dc} (V)	V_{ac} (V)	R.F.(γ) (Practical) (V_{ac}/V_{dc})	R.F.(γ) (Theoretical) ($1/ 4\sqrt{3}fCR_L$)

Voltage Regulation: -

Condition	Output Voltage (V)
No Load (VNL)	
Full Load (VFL)	

Calculation: -

Voltage Regulation = [(VNL − VFL)/VFL] × 100 = _____ %

Teacher's Sign and Date